Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Equilibrium

Meaning in my words How would you explain equilibrium to a friend?	**How can you tell?** A cart rolls straight at steady speed. What must the horizontal forces add up to?
Example or sketch Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left.	**Non-example and why** A tossed ball is briefly still at the top. Is it in equilibrium there?

Static and dynamic equilibrium

Meaning in my words What do static and dynamic equilibrium have in common? How do they differ?	**How can you tell?** A parked cart and a cart rolling straight at steady speed both have zero net force. How do they differ?
Example or sketch Draw equal-time dots for each cart.	**Non-example and why** A bike turns at steady speed. Would that be dynamic equilibrium?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Missing force

Meaning in my words How would you explain missing force to a friend?	**How can you tell?** A cart is in equilibrium with 7 N right. If there is only one other horizontal force, what must it be?
Example or sketch Two cables supply 3 N and 5 N upward on a stationary object. What downward force completes the diagram?	**Non-example and why** A partner adds another upward 8 N. What would happen to the vertical force sum?

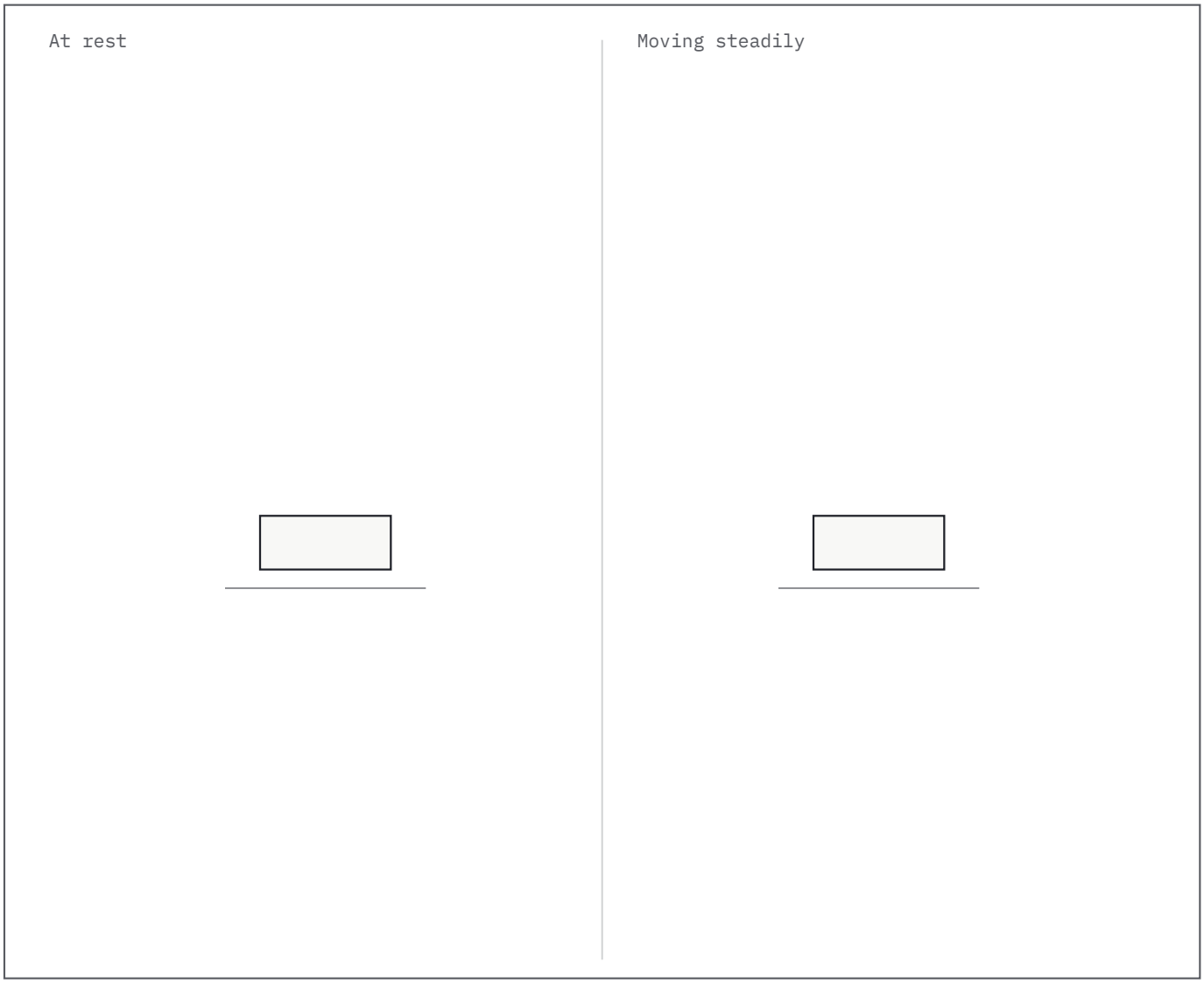
Instantaneous rest

Meaning in my words How would you explain instantaneous rest to a friend?	**How can you tell?** At the top of a ball's straight upward toss, what is zero and what is still acting?
Example or sketch How would you draw velocity and acceleration at that instant?	**Non-example and why** "It stopped, so the forces must balance." What does the next instant show?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.



My marks mean

Stopped or coasting?

18

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Distinguish zero acceleration from zero velocity.

CHOOSE ONE JOB TOGETHER

FLIGHT CONTROLLER ☐

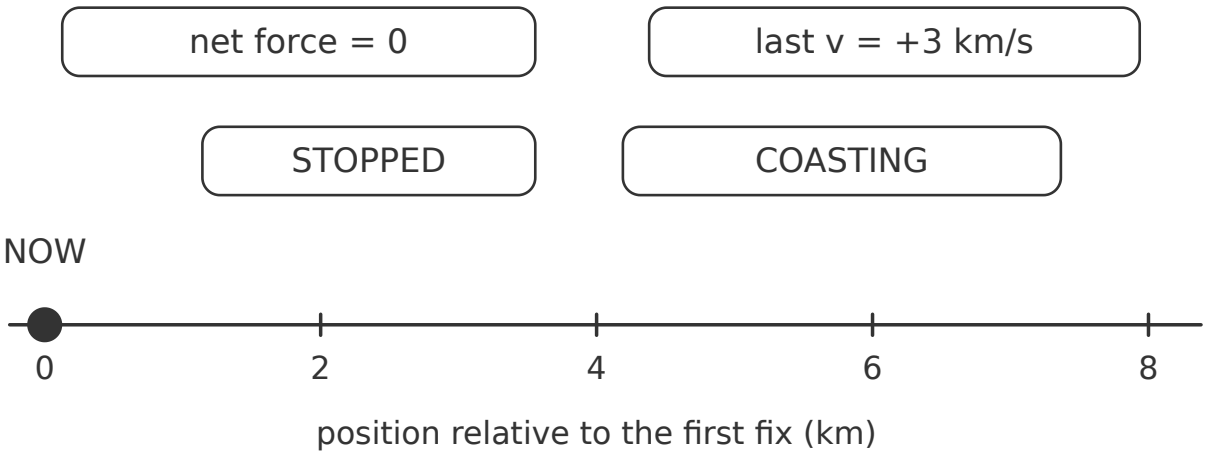
Choose the status graphic. Mark where XJ should be two seconds later relative to the first sighting.

BROADCAST PRODUCER ☐

Choose the on-air label. Explain why zero acceleration and zero speed do not mean the same thing.

Point to a clue. Keep your first marks; add the update using the color key.

The flight display reads "net force zero." XJ's last velocity was **+3 km/s**. Two graphics say STOPPED and COASTING.



01 OUR FIRST CALL

Which status should go live? Use a second position to support your call.

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Can the old status describe the next interval? Explain what the new report changes.

Switch speakers. Does zero net force mean zero speed?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks